

Mechanisms of antimicrobial resistance in bacteria.



The treatment of bacterial infections is increasingly complicated by the ability of bacteria to develop resistance to antimicrobial agents. Antimicrobial agents are often categorized according to their principal mechanism of action. Mechanisms include interference with cell wall synthesis (e.g., beta-lactams and glycopeptide agents), inhibition of protein synthesis (macrolides and tetracyclines), interference with nucleic acid synthesis (fluoroquinolones and rifampin), inhibition of a metabolic pathway (trimethoprim-sulfamethoxazole), and disruption of bacterial membrane structure (polymyxins and daptomycin). Bacteria may be intrinsically resistant to > or =1 class of antimicrobial agents, or may acquire resistance by de novo mutation or via the acquisition of resistance genes from other organisms. Acquired resistance genes may enable a bacterium to produce enzymes that destroy the antibacterial drug, to express efflux systems that prevent the drug from reaching its intracellular target, to modify the drug's target site, or to produce an alternative metabolic pathway that bypasses the action of the drug. Acquisition of new genetic material by antimicrobial-susceptible bacteria from resistant strains of bacteria may occur through conjugation, transformation, or transduction, with transposons often facilitating the incorporation of the multiple resistance genes into the host's genome or plasmids. Use of antibacterial agents creates selective pressure for the emergence of resistant strains. Herein 3 case histories-one involving *Escherichia coli* resistance to third-generation cephalosporins, another focusing on the emergence of vancomycin-resistant *Staphylococcus aureus*, and a third detailing multidrug resistance in *Pseudomonas aeruginosa*--are reviewed to illustrate the varied ways in which resistant bacteria develop.